Section I Number 1 and 3

Bryan Dion Cellona and Nathalie Faith Ricafort

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Item 1: Probability of a Defective Product

A company has three factories producing a product. The probability that a randomly selected product is defective is calculated using the production proportions and defective rates.

```
# Production proportions
# These example values satisfy the required conditions
x1 <- 0.30
x2 <- 0.35
x3 <- 0.35

# Defective rates
y1 <- 0.03
y2 <- 0.04
y3 <- 0.05

# Check the production proportions
stopifnot(
  all(c(x1, x2, x3) >= 0.10),
  all(c(x1, x2, x3) <= 0.40),
  abs(sum(x1, x2, x3) - 1) < 1e-8
)

# Check the defective rates
stopifnot(
  all(c(y1, y2, y3) >= 0.01),
  all(c(y1, y2, y3) <= 0.05),
  abs(sum(y1, y2, y3) - 0.12) < 1e-8
)

# Calculate the overall probability of a defective product
prob_defective <- x1*y1 + x2*y2 + x3*y3

# Display the results
cat(
  "Probability that a product is defective:",
  sprintf("%.4f", prob_defective), "\n"
)
```

```
## Probability that a product is defective: 0.0405
```

```r
cat(
  "Percentage defective:",
  sprintf("%.2f%%", prob_defective * 100), "\n"
)
```

```
## Percentage defective: 4.05%
```

Item 3: Geometric Distribution Simulation

Simulate 10,000 searches until the first success, assuming each site has a 60% chance of containing the key phrase.

```r
# Set the probability of success
p <- 0.60

# Number of simulations
N <- 10000

# Set a seed for reproducibility
set.seed(1110)

# Generate searches until the first success
# rgeom() counts failures, so add 1 to count total searches
X <- rgeom(N, prob = p) + 1

# Calculate the mean and variance
mean_X <- mean(X)
var_X <- var(X)

# Display the results
cat("COMPLETE SIMULATION\n")
```

```
## COMPLETE SIMULATION
```

```r
cat(sprintf("Mean: %.4f\n", mean_X))
```

```
## Mean: 1.6688
```

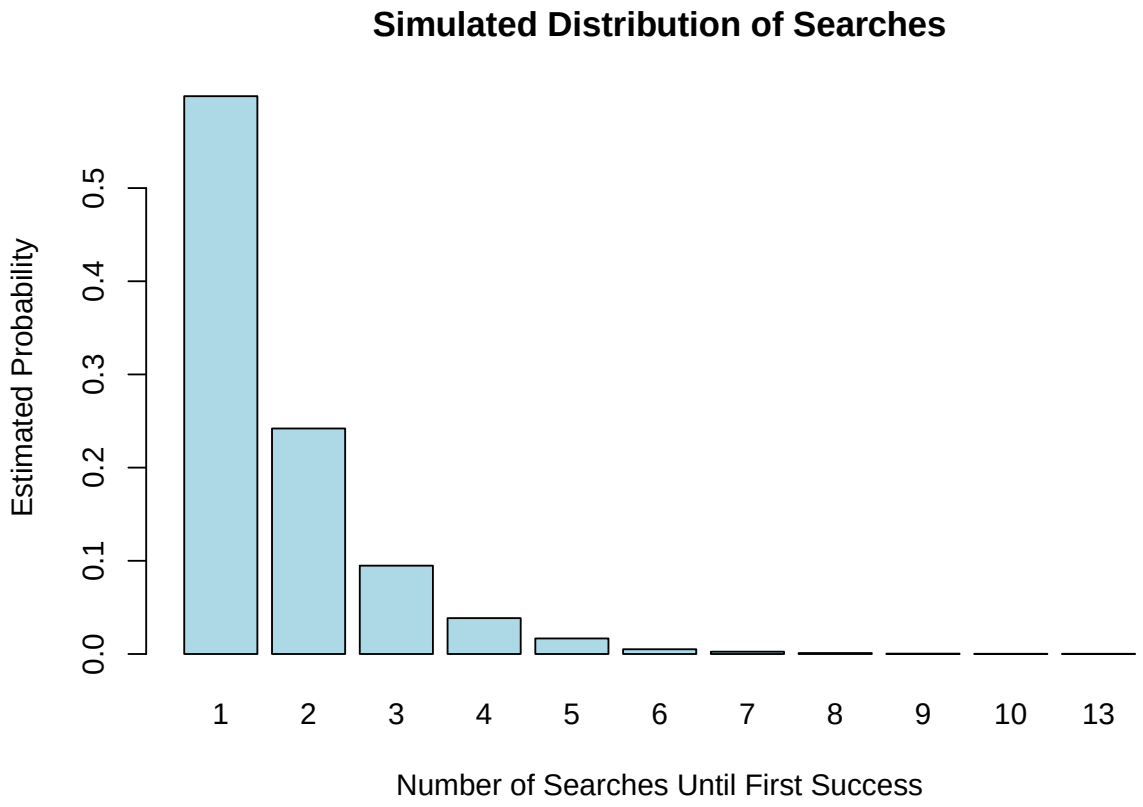
```r
cat(sprintf("Variance: %.4f\n", var_X))
```

```
## Variance: 1.1248
```

Simulated Probability Distribution

```r
# Estimate the probability of each number of searches
simulated_pmf <- prop.table(table(X))
```

```r
# Plot the estimated probability distribution
barplot(
  simulated_pmf,
  main = "Simulated Distribution of Searches",
  xlab = "Number of Searches Until First Success",
  ylab = "Estimated Probability",
  col = "lightblue",
  border = "black"
)
```



Conditional Distribution After Three Failures

```r
# Select simulations where more than 3 searches were required
# Subtract 3 to count the additional searches after the first
# three unsuccessful searches
X_conditional <- X[X > 3] - 3

# Calculate conditional mean and variance
mean_conditional <- mean(X_conditional)
var_conditional <- var(X_conditional)

# Display results
cat("CONDITIONAL SIMULATION AFTER THREE FAILURES\n")
```

```
## CONDITIONAL SIMULATION AFTER THREE FAILURES
```

```
cat("Number of qualifying simulations:",
    length(X_conditional), "\n")
```

```
## Number of qualifying simulations: 646
```

```
cat(sprintf("Conditional mean: %.4f\n", mean_conditional))
```

```
## Conditional mean: 1.6718
```

```
cat(sprintf("Conditional variance: %.4f\n", var_conditional))
```

```
## Conditional variance: 1.2379
```

Check the Memoryless Property

```
# Estimate P(X = 4 | X > 3) and P(X = 5 | X > 3)
prob_X4_given_Xgt3 <- sum(X == 4) / sum(X > 3)
prob_X5_given_Xgt3 <- sum(X == 5) / sum(X > 3)

# Estimate P(X = 1) and P(X = 2)
prob_X1 <- mean(X == 1)
prob_X2 <- mean(X == 2)

# Display the comparisons
cat("MEMORYLESS PROPERTY COMPARISON\n")
```

```
## MEMORYLESS PROPERTY COMPARISON
```

```
cat(sprintf("P(X = 4 | X > 3): %.4f\n",
            prob_X4_given_Xgt3))
```

```
## P(X = 4 | X > 3): 0.5960
```

```
cat(sprintf("P(X = 1):         %.4f\n", prob_X1))
```

```
## P(X = 1):         0.5986
```

```
cat(sprintf("P(X = 5 | X > 3): %.4f\n",
            prob_X5_given_Xgt3))
```

```
## P(X = 5 | X > 3): 0.2585
```

```r
cat(sprintf("P(X = 2):          %.4f\n", prob_X2))
```

```
## P(X = 2):          0.2420
```

Interpretation

The geometric distribution has the memoryless property. The probability distribution of the additional searches after three unsuccessful searches is the same as the distribution at the beginning of the search.

The simulated probabilities and statistics may differ slightly because they are estimates based on 10,000 simulations. “‘